

Comparison between “Audiovisual Eyeglasses Distraction” and “Biofeedback Therapy” as Behavior Guidance Tools in Managing Pediatric Dental Patients

Ramakrishna Juvva¹, Vasa AA Kumar², Madu G Prasad³, Suzan Sahana⁴, KVK Santosh Kumar⁵, Snigdha Gavini⁶

Received on: 14 February 2025; Accepted on: 06 June 2025; Published on: 14 October 2025

ABSTRACT

Aims and background: Children fail to seek dental treatment for numerous reasons, one significant among them being anxiety. Though multiple strategies are proposed to minimize anxiety, noninvasive techniques have a higher acceptance level. Distraction with audiovisual eyeglasses (AVE) has gained acceptance as effective in the medical field.

Methods: Study participants included 60 children aged 6–7 without dental experience and requiring similar restorative procedures. The children were randomly allocated, 20 each, to a group according to the technique employed while undergoing the treatment: group A—AVE, group B—biofeedback therapy, and group C—the control group. The children's experience with each type of treatment was noted before, during, and after the procedure using the parameters of pain measurement, pulse rate, and oxygen saturation.

Results: The Kruskal–Wallis test was done to compare the oxygen saturation rate. The analysis of variance (ANOVA) and Tukey honest significant difference (HSD) test were done to compare the pulse rate. The pain rate was compared using the Mann–Whitney test. No significant difference was observed between groups A and B, but a statistically significant difference was seen between groups A and C and B and C.

Conclusion: AVE and biofeedback therapy as behavior guidance tools were found to be equally effective in alleviating anxiety in children.

Keywords: Audiovisual aids, Biofeedback therapy, Dental anxiety.

Journal of South Asian Association of Pediatric Dentistry (2025): 10.5005/jp-journals-10077-3357

INTRODUCTION

Dental anxiety is a significant barrier that may prevent children from seeking necessary dental care, often due to fear of pain and invasive procedures. This anxiety can manifest in varying degrees, from mild tension to severe distress, potentially leading to total avoidance of treatment.^{1,2} Effective management of anxiety is crucial for ensuring successful pediatric dental care.¹ Various pharmacologic and nonpharmacologic methods have been explored to achieve cooperative behavior in children.^{1,3} Behavioral techniques, such as verbal reassurance, tell-show-do, physical contact, and relaxation exercises, are beneficial but may not always be effective, especially for highly uncooperative patients.³

Recent advancements, including audiovisual eyeglasses (AVE) and biofeedback therapy, offer promising alternatives for anxiety management in pediatric dentistry. AVE provides both auditory and visual stimulation, engaging children without interfering with dentist–patient communication.⁴ Biofeedback therapy is a self-regulation technique that allows individuals to control physiological processes such as heart rate and breathing patterns, ultimately reducing stress and anxiety.¹ However, the success of these techniques requires further investigation. The present study aims to evaluate the efficacy of AVE distraction and biofeedback therapy in managing pediatric dental anxiety during restorative procedures, to enhance anxiety management strategies in clinical practice.

METHODS

This study was conducted on children attending the outpatient department of pediatric and preventive dentistry, all in good general

^{1–4}Department of Pediatric and Preventive Dentistry, St Joseph Dental College, Eluru, Andhra Pradesh, India

⁵Department of Pediatric and Preventive Dentistry, Anil Neerukonda Institute of Dental Sciences, Visakhapatnam, Andhra Pradesh, India

⁶Department of Conservative Dentistry and Endodontics, St Joseph Dental College, Eluru, Andhra Pradesh, India

Corresponding Author: Suzan Sahana, Department of Pediatric and Preventive Dentistry, St Joseph Dental College, Eluru, Andhra Pradesh, India, Phone: +91 9886231079, e-mail: drsuzansahana@gmail.com

How to cite this article: Juvva R, Vasa AAK, Prasad MG, *et al.* Comparison between “Audiovisual Eyeglasses Distraction” and “Biofeedback Therapy” as Behavior Guidance Tools in Managing Pediatric Dental Patients. *J South Asian Assoc Pediatr Dent* 2025;8(3):205–209.

Source of support: Nil

Conflict of interest: None

health with no systemic illnesses or history of hospitalization. Before the study commenced, informed consent was obtained from parents/guardians. The Institutional Ethics Committee granted an ethical clearance approving the study protocol (Serial No. CEC/18/2014).

The sample size was calculated based on previous studies^{1,2} with 95% power. The estimated sample size for the current study was 60. The participants were recruited between June and December 2015 and completed the study over 6 months.

A single-blinded, randomized controlled trial was designed to evaluate the effectiveness of AVE distraction and biofeedback therapy. About 60 children (30 boys and 30 girls), aged 6–7 years, with no prior dental experience and requiring similar restorative

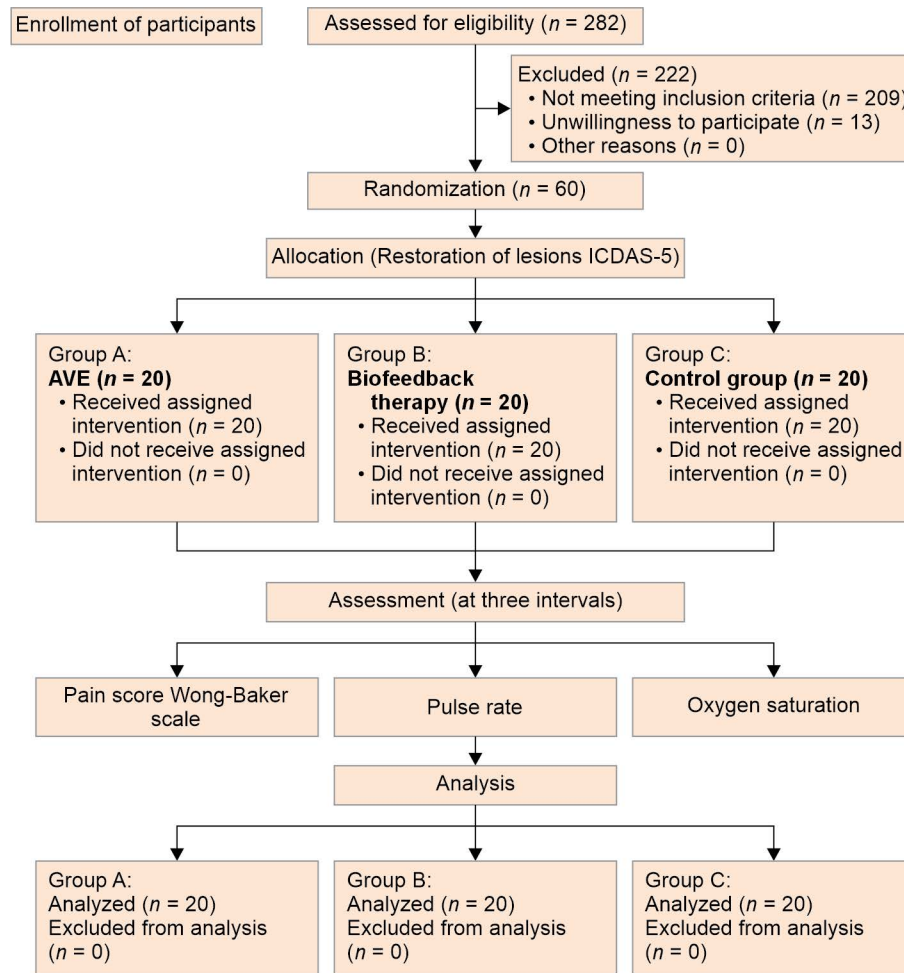


Fig. 3: Consort flowchart of the research

Table 1: Intergroup comparison of oxygen saturation variables

Time	p-value*	Significance
Before	0.367	Not significant
During	0.529	Not significant
After	0.539	Not significant

*p-value < 0.05 statistically significant; Kruskal–Wallis test—asymptomatic significances are displayed

Table 2: Pair-wise comparison of pulse rates before, during, and after treatment

Comparative groups	Before p-value*	During p-value*	After p-value*
Groups A vs B	0.272	0.998	0.482
Groups A vs C	0.803	0.010*	0.022*
Groups B vs C	0.081	0.008*	0.001*

*p-value < 0.05 significant

- Pulse rate: No significant differences in the baseline pulse rate were noted before treatment. However, the observations revealed significant differences between groups A vs C and groups B vs C during and after treatment ($p < 0.05$) (Table 2).
- Pain ratings: A statistically significant difference was noted in pain scores between groups A vs C and B vs C during and after treatment (Table 3).

- Gender differences: No significant difference was noted between males and females for the measured variables across all the groups (Table 4).

DISCUSSION

Behavior management is crucial in dentistry for children to ensure effective treatment and cooperation.⁷ Disruptive behavior in the dental operatory increases treatment difficulty, duration, and the risk of injury.⁸ Anxiety during the first dental visit can amplify pain perception, necessitating interventions that enhance patient comfort. This study assessed AVE and biofeedback therapy as anxiety-reducing techniques among children aged 6–7, minimizing behavioral disparities due to developmental stages.

Caries diagnosis using ICDAS criteria for lesion detection ensured a standardized approach.⁵ Dental anxiety is a dispositional factor affecting treatment experiences and pain perception.⁸ This study utilized the Wong–Baker faces pain scale, a validated tool for assessing self-reported pain in school-aged children.⁶ Physiological parameters such as pulse rate and oxygen saturation were used as objective stress indicators.^{9,10}

Results indicated that both AVE and biofeedback therapy significantly reduced anxiety compared to the control group, aligning with prior research.^{11,12} Significant reductions in pulse rate were observed in the AVE and biofeedback groups during and

Table 3: Intergroup comparison of pain scale scores

Groups	Time	Mann–Whitney	Wilcoxon W	p-value*
Groups A vs B	Before	135.500	345.500	0.075
	During	193.500	403.500	0.0854
	After	168.500	378.500	0.244
Groups A vs C	Before	172.00	382.000	0.442
	During	66.000	276.000	0.000*
	After	43.500	253.500	0.000*
Groups B vs C	Before	107.000	317.000	0.011*
	During	85.500	295.500	0.001*
	After	88.500	298.500	0.002*

*p-value < 0.05 significant

Table 4: Comparison of all variables between genders in all three groups

Variable	Group A p-value			Group B p-value			Group C p-value		
	Before	During	After	Before	During	After	Before	During	After
O ₂ saturation	0.739	0.481	0.529	0.529	0.280	0.912	0.853	0.853	0.631
Pulse rate	0.393	0.739	0.912	0.796	0.739	0.853	0.631	0.796	0.481
Pain rating	0.393	1.000	0.796	0.631	0.393	0.436	0.481	0.281	0.912

Mann–Whitney U test—asymptomatic results were displayed

after treatment, confirming the efficacy of both techniques.^{3,4,13–17} The distraction provided by AVE likely helped children focus on engaging content rather than the procedure. Increased anxiety among children in the control group may be due to sensory stimuli from dental instruments.

Biofeedback therapy also demonstrated effectiveness, as children learned to regulate their physiological responses through relaxation techniques. However, biofeedback requires cognitive effort, motivation, and sustained attention, limiting its feasibility for younger children.

The oxygen saturation levels did not differ significantly among groups, indicating that all participants maintained normal oxygenation throughout treatment, consistent with previous studies.^{18,19} Gender-wise, no observable difference was noted for all parameters, aligning with prior research findings.^{14,20}

Limitations of AVE included improper fit for smaller faces and sterilization concerns; this was managed using disposable speaker covers. Cost considerations may also impact its widespread use. Biofeedback therapy, while effective, requires extensive training and reinforcement.

CONCLUSION

- Audiovisual eyeglasses and biofeedback therapy are effective, noninvasive behavioral management tools for reducing pediatric dental anxiety.
- Both techniques demonstrated similar efficacy in alleviating anxiety during restorative treatment.
- No significant gender-based differences were observed in behavioral responses.

Further randomized controlled trials across diverse age-groups and dental procedures are needed to validate these findings.

Clinical Significance

The integration of innovative, noninvasive anxiety management strategies in pediatric dental operatories enhances patient cooperation, ensuring positive dental experiences and long-term oral health benefits.

ORCID

Vasa AA Kumar  <https://orcid.org/0000-0002-2161-6723>

Suzan Sahana  <https://orcid.org/0000-0003-0774-307X>

REFERENCES

1. Dedeepya P, Nuvvula S, Kamatham R, et al. Behavioural and physiological outcomes of biofeedback therapy on dental anxiety of children undergoing restorations: a randomized controlled trial. *Eur Arch Paediatr Dent* 2014;15(2):97–103. DOI: 10.1007/s40368-013-0070-3
2. Jindal R, Kaur R. Can we tune our paediatric patients? *Int J Clin Pediatr Dent* 2011;4(3):186–189. DOI: 10.5005/jp-journals-10005-1107
3. Ram D, Shapira J, Holan G, et al. Audiovisual video eyeglass distraction during dental treatment in children. *Quintessence Int* 2010;41(8):673–679. PMID: 20657857.
4. Florella M, Cohen SE, Ram RD. Audiovisual iatrosedation with video eyeglasses distraction method in pediatric dentistry: case history. *J Int Dent Med Res* 2010;3(3):133–136.
5. Ismail AI, Sohn W, Tellez M, et al. The international caries detection and assessment system (ICDAS): an integrated system for measuring dental caries. *Community Dent Oral Epidemiol* 2007;35:170–178. DOI: 10.1111/j.1600-0528.2007.00347.x
6. Khatri A, Kalra N. A comparison of two pain scales in the assessment of dental pain in East Delhi children. *ISRN Dent* 2012;2012:247351. DOI: 10.5402/2012/247351
7. Roberts JF, Curzon MEJ, Koch G, et al. Review: behaviour management techniques in paediatric dentistry. *Eur Arch Pediatr Dent* 2010;11(4):166–174. DOI: 10.1007/BF03262738

8. Brill WA. Child behavior in a private pediatric dental practice associated with types of visits, age and socio-economic factors. *J Clin Pediatr Dent* 2000;25(1):1–7. PMID: 11314346.
9. Sowjanya V, Tandon S, Tharian E. Physiological response to dental anxiety in children. *J Indian Soc Pedod Prev Dent* 1995;13:13–17. PMID: 9522733.
10. Rayen R, Muthu MS, Rao RC, et al. Evaluation of physiological and behavioral measures in relation to dental anxiety during sequential dental visits in children. *Indian J Dent Res* 2006;17(1):27–34. DOI: 10.4103/0970-9290.29895
11. Rodolfa ER, Kraft W, Reilley RR. Etiology and treatment of dental anxiety and phobia. *Am J Clin Hypn* 1990;33:22–28. DOI: 10.1080/00029157.1990.10402897
12. Alwin NP, Murray JJ, Britton PG. An assessment of dental anxiety in children. *Br Dent J* 1991;171:201–207. DOI: 10.1038/sj.bdj.4807661
13. Hoge MA, Howard MR, Wallace DP, et al. Use of video eyewear to manage distress in children during restorative dental treatment. *Pediatr Dent* 2012;34(5):378–382. PMID: 23211912.
14. Nuvvula S, Alahari S, Kamatham R, et al. Effect of audiovisual distraction with 3D video glasses on dental anxiety of children experiencing administration of local analgesia: a randomized clinical trial. *Eur Arch Paediatr Dent* 2015;16:43–50. DOI: 10.1007/s40368-014-0145-9
15. Asvanund Y, Mitrakul K, Juhong RO, et al. Effect of audiovisual eyeglasses during local anesthesia injections in 5- to 8-year-old children. *Quintessence Int* 2015;46(6):513–521. DOI: 10.3290/j.qi.a33932
16. Kaur R, Jindal R, Dua R, et al. Comparative evaluation of the effectiveness of audio and audiovisual distraction aids in the management of anxious paediatric dental patients. *J Indian Soc Pedod Prev Dent* 2015;33(3):192–203. DOI: 10.4103/0970-4388.160357
17. Custódio NB, Costa FDS, Cademartori MG, et al. Effectiveness of virtual reality glasses as a distraction for children during dental care. *Pediatr Dent* 2020;42(2):93–102. PMID: 32276674.
18. Yelderman M, New W Jr. Evaluation of pulse oximetry. *Anesthesiology* 1983;59:349–352. DOI: 10.1097/00000542-198310000-00015
19. Prabhakar AR, Marwah N, Raju OS. A comparison between audio and audiovisual distraction techniques in managing anxious pediatric dental patients. *J Indian Soc Pedod Prev Dent* 2007;25(4):177–182. DOI: 10.4103/0970-4388.37014
20. El-Sharkawi HFA, El-Housseiny AA, Aly AM. Effectiveness of new distraction technique on pain associated with injection of local anesthesia for children. *Pediatr Dent* 2012;34(2):e35–e38. PMID: 22583875.